# Tutorial 4 – Operators and Expressions

**1.** Which of the following arithmetic expressions are valid? If valid calculate the value of the expression, otherwise give reason why the expression is invalid.

a. 17 % 5.0 + 9 % 4
b. -22 % 3
c. (55 / 3) / 2 + 5 % 2

**2.** Determine the value of the following expressions where, ***a = 2, b = 3, and c = 4.***

a. (a<b) || (b<c)
b. !(a>b) || (c<=b)
c. (a>b) || (b<c) && (a<b) || (a<c)
**d.** ((a>b) || (b<c)) && (a>b)

3. Calculate the value of the following assignment statements, where ***a = 10*** and ***a, b, c*** and ***answer*** are *integer variables.*

a. answer = a + (b = 10)
b. answer = a + (b = 10) – 8
c. answer = a + (b = (c = 12)) - 8;

4. Writing numeric expressions in C++ involves a straightforward translation of an arithmetic expression using C++ operators. Re-write the following arithmetic expression using C++ operators and display the answer where all the variables = 1.
(output 247.4)

$$\frac{3 + 4x}{5} - \frac{10(y - 5)(a + b + c)}{x} + 9\left(\frac{4}{x} + \frac{9 + x}{y}\right)$$

5. Write a C++ program to calculate the value of z, where $z = \frac{\sqrt{5x^5 - (3x^2)/y^3}}{2x}$ instruct the user to enter the value of x as a positive integer value (between 2 to 5) and value of y as 2. Use mathematical functions to calculate the square root and power of the values.

**Sample outputs**
Enter any number between 2 to 5: **3**
Z = 5.80140

**Using condition ? result1 : result2**

6. Write a C++ program to enter a value from the keyboard. Check whether the input value is even or odd and display the number and "is an even number" if it is even, otherwise display the number and "is an odd number".

**Sample outputs**

Enter a number: 3
3 is an odd number

Enter a number: 14
14 is an even number

7. Create an admission criteria checker
   Eligibility = math ≥ 60 AND (science ≥ 60 OR english ≥ 60)

8. Write a C++ program that calculates a discount based on the total amount. If the amount is greater than 2000, apply a 10% discount; otherwise, apply a 5% discount. Then, display the discount.

   Example: Amount 2500.0
   Discount: 250